AML / DL COURSE · PROJECT 5

forecastkit

An open toolkit for probabilistic demand forecasting: Kalman filters, LightGBM and global DeepAR, compared on M5 under one leak-proof protocol

DJ · Inventory Demand Forecasting via Kalman Filtering and Global DeepAR Models

THE PROBLEM

Ordering stock is a bet on a range, not a number

Quantiles, not points

The 90th percentile of demand over the lead time is the order-up-to level for a 90% service target.

Sparse, censored data

Most retail series are intermittent, and a zero can mean no demand or an empty shelf.

New items, little history

A model that needs a long history fails on launches. Global models can borrow strength from similar items.

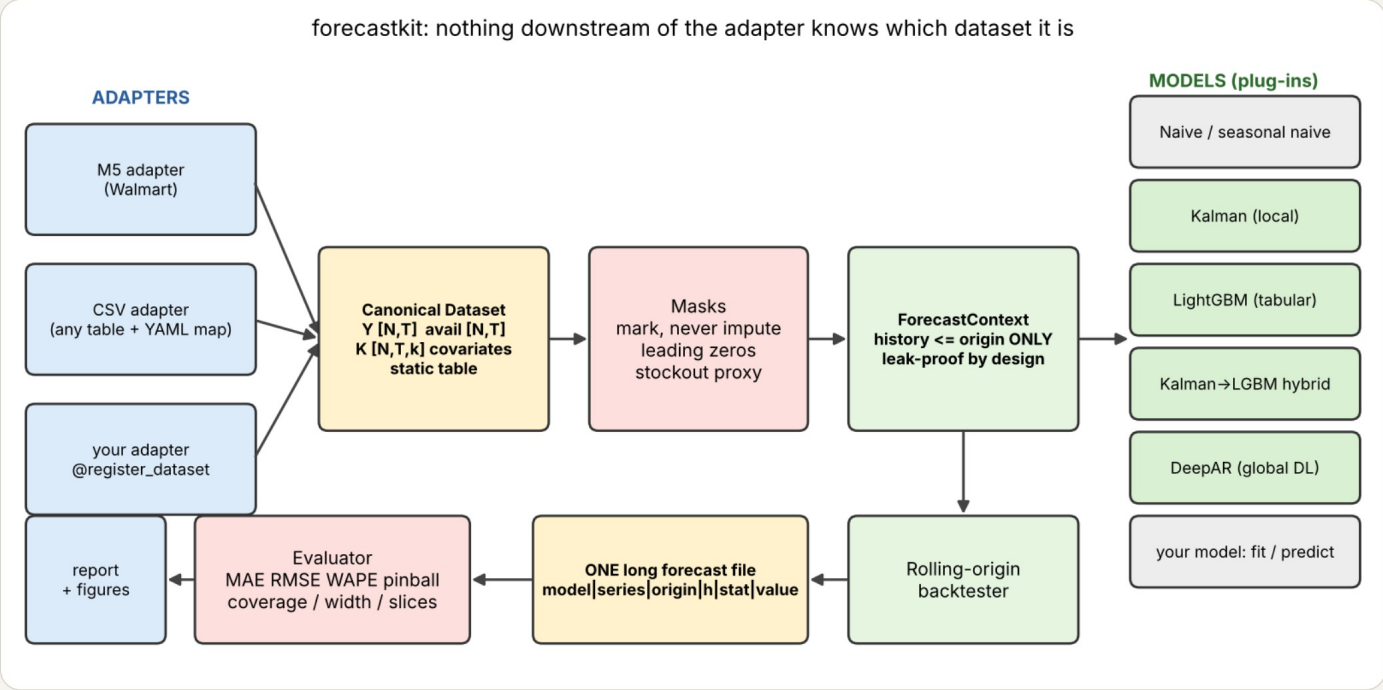
Question: when does a global deep model beat a local filter or a tabular baseline, and how well calibrated is each?

Data and experiment design

Item	Setting
Data	M5 Walmart daily unit sales, with calendar, events, SNAP and prices
Series	About 1,200: 3 stores × ~400 items, frozen and stratified by department and demand class
Horizon	28 days, quantiles 0.05 to 0.95
Origins	4 rolling test origins, plus 1 earlier validation origin for tuning only
Models	Seasonal naive, Kalman per SKU, LightGBM, Kalman→LightGBM, DeepAR

Every model sees the same context, the same series, the same origins and the same scoring file.

forecastkit: two contracts, everything else plugs in



Any dataset becomes Y, avail, K, static through a small adapter.

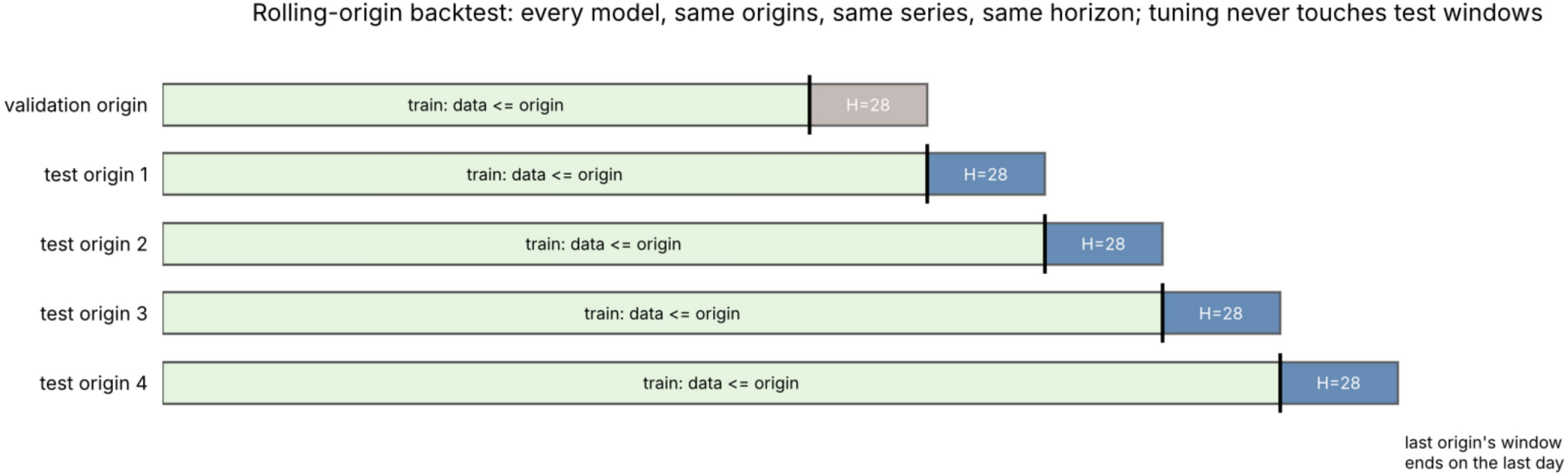
Any model implements fit and predict.

One forecast file is all the evaluator reads.

Proven on M5-format data and on an unrelated bakery CSV with only the adapter block changed.

FAIR PROTOCOL

Rolling origins, no peeking



Tests perturb every future target and check that no model's forecast changes.

STOCKOUTS, COLD START, UNCERTAINTY

Mark what we do not know; never impute it

Stockout proxy

A long run of zeros, judged against the item's own sales rhythm, is flagged as censored. It is a proxy, never ground truth.

Fair scoring

Flagged days are missing data for training. Results are reported on all days and on non-censored days.

Simulated cold start

About 7% of series reveal only their last 28 days. Global models use an unknown-item embedding.

Honest intervals

Seven quantiles per forecast, scored by pinball loss, coverage and calibration.

Three families, one interface

LOCAL

Kalman per SKU

Local level plus stochastic weekly seasonal on a square-root scale. Skips censored days. A diffuse prior gives wide intervals on short histories.

TABULAR

LightGBM

Direct by horizon bucket, Poisson mean plus one quantile model per level, strictly causal lag and rolling features. The hybrid adds Kalman state as features.

GLOBAL DEEP

DeepAR

One LSTM for all series, negative binomial likelihood, item embeddings, loss masking, quantiles from 200 sampled paths.

Baselines: naive, seasonal naive, moving average. Post-hoc ensemble: quantile average of global and local.

OPEN, NOT RIGID

Add a model or a dataset in about ten lines

New model

```
@register_model("mine")
class Mine(BaseForecaster):
  def fit(self, ctx): ...
  def predict(self, quantiles):
    return self.finalize(q, mean=m)
```

New dataset

```
@register_dataset("my_shop")
def load(path):
  ... build Y, avail, K, static
  return Dataset(...)
```

Or skip code: a YAML column map with type: csv.

Then point the YAML config at it, by name or by import path. The evaluator is untouched.

How every model is scored

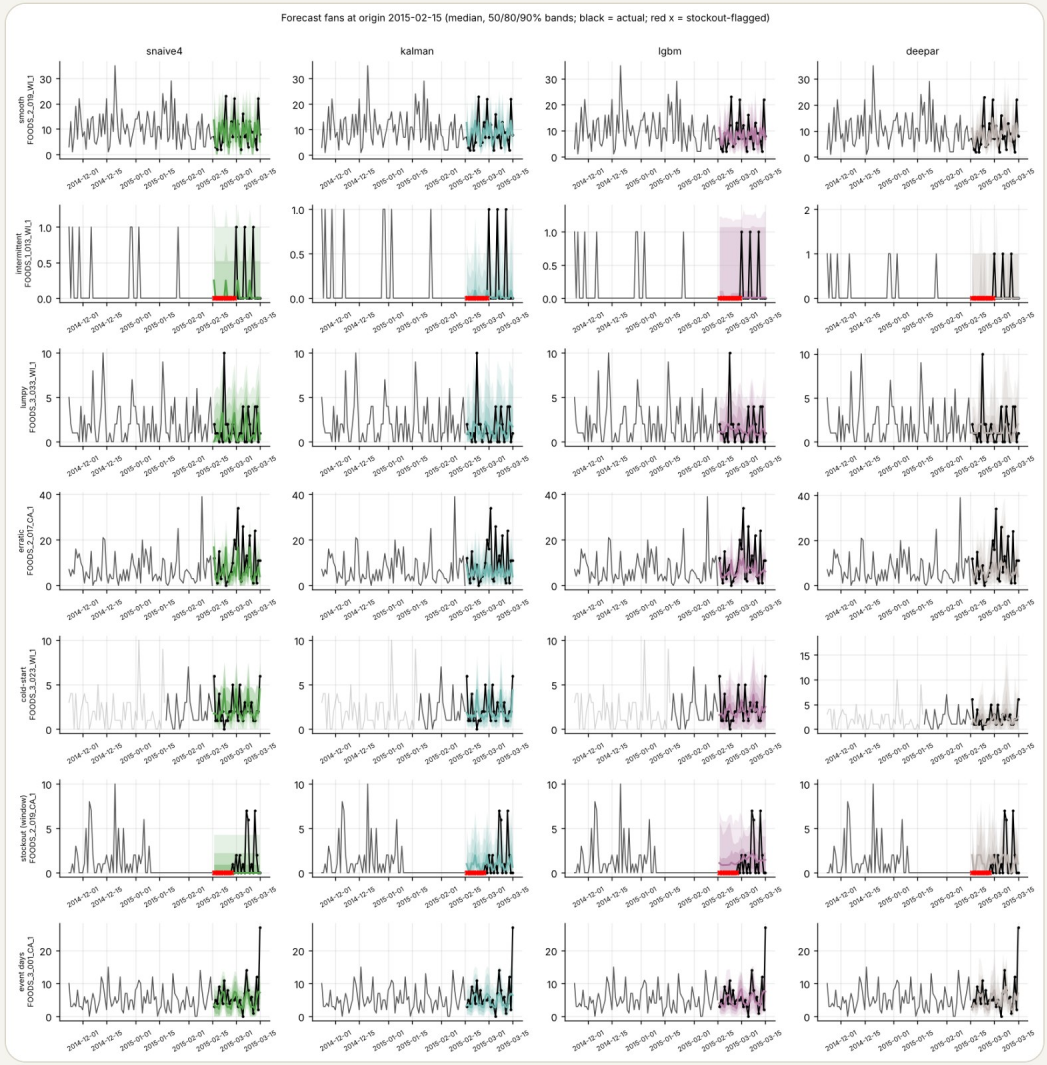
Metric	What it tells us
MAE, RMSE	Typical and large-miss error of the median and the mean forecast
WAPE	Total absolute error over total sales, aggregated so big sellers count in proportion
Pinball loss	Accuracy of the whole quantile set, averaged over seven levels
Coverage, width	Do 80% and 90% intervals contain that share of actuals, and how wide are they
Slices	By horizon, origin, demand class, volume tier, cold start, censored or not
Significance	Paired bootstrap over series against the seasonal naive reference

Calibration for counts is shown as a band, because zero has probability mass.

M5 results: [M5 RESULT headline]

Model	MAE	WAPE	Pinball	Cov90	Time
Seasonal naive	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]
Kalman per SKU	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]
LightGBM	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]
Kalman→LightGBM	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]
DeepAR	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]
Ensemble	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]	[M5 RESULT]

Placeholders: fill from outputs/m5_main/report/RESULTS.md after the real M5 run. No numbers on this slide are measured yet.



SYNTHETIC DATA, NOT M5

What one auto-generated report figure looks like

Rows are series types: smooth, intermittent, lumpy, erratic, cold start, stockout window, event days. Columns are models. Bands are the 50, 80 and 90% intervals; red marks are stockout-flagged days.

This is the pipeline's own output on simulated M5-format data. The real M5 figure replaces it after the run.

Ablations: what drives each result

Study	Variants	Question
DeepAR	Item embedding, price covariates, likelihood, context length, loss masking	What the global model needs
Stockout	28-day rule, 14-day rule, no masking	Does censoring help, and how fragile is the rule
Kalman	Square-root vs raw, shared vs per-series noise, window	Which local design choices matter
LightGBM	Poisson vs Tweedie vs L2, price features, bucket size	How much is feature engineering

Each is one YAML file; tables show all days and non-censored days. Effects: [M5 RESULT].

Limitations we state up front

- Stockouts are inferred from zeros; M5 has no inventory records.
- Cold start is simulated by hiding history, not observed.
- One retailer and about 1,200 series, so results may not transfer.
- The Kalman model is Gaussian on a transformed scale, weak on very intermittent items.
- DeepAR is a compact in-repo version; the GPU path is untested.

What you can take away and reuse

A working toolkit

pip-installable, YAML-driven, 35 passing tests

A fair comparison

Leak-proof origins, masked stockouts, calibrated intervals

A repeatable report

One command writes tables and figures from the forecast file

Next: run on real M5, fill the [M5 RESULT] slots, then try a new dataset.